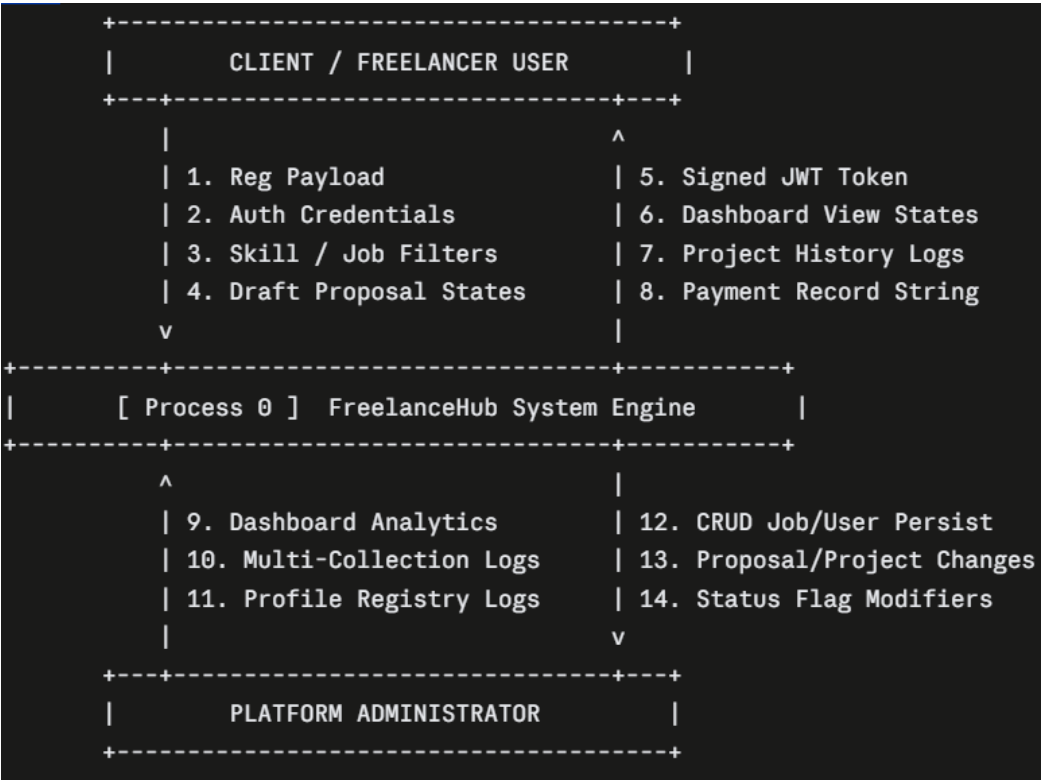


Data Flow Diagram Architectural Visuals

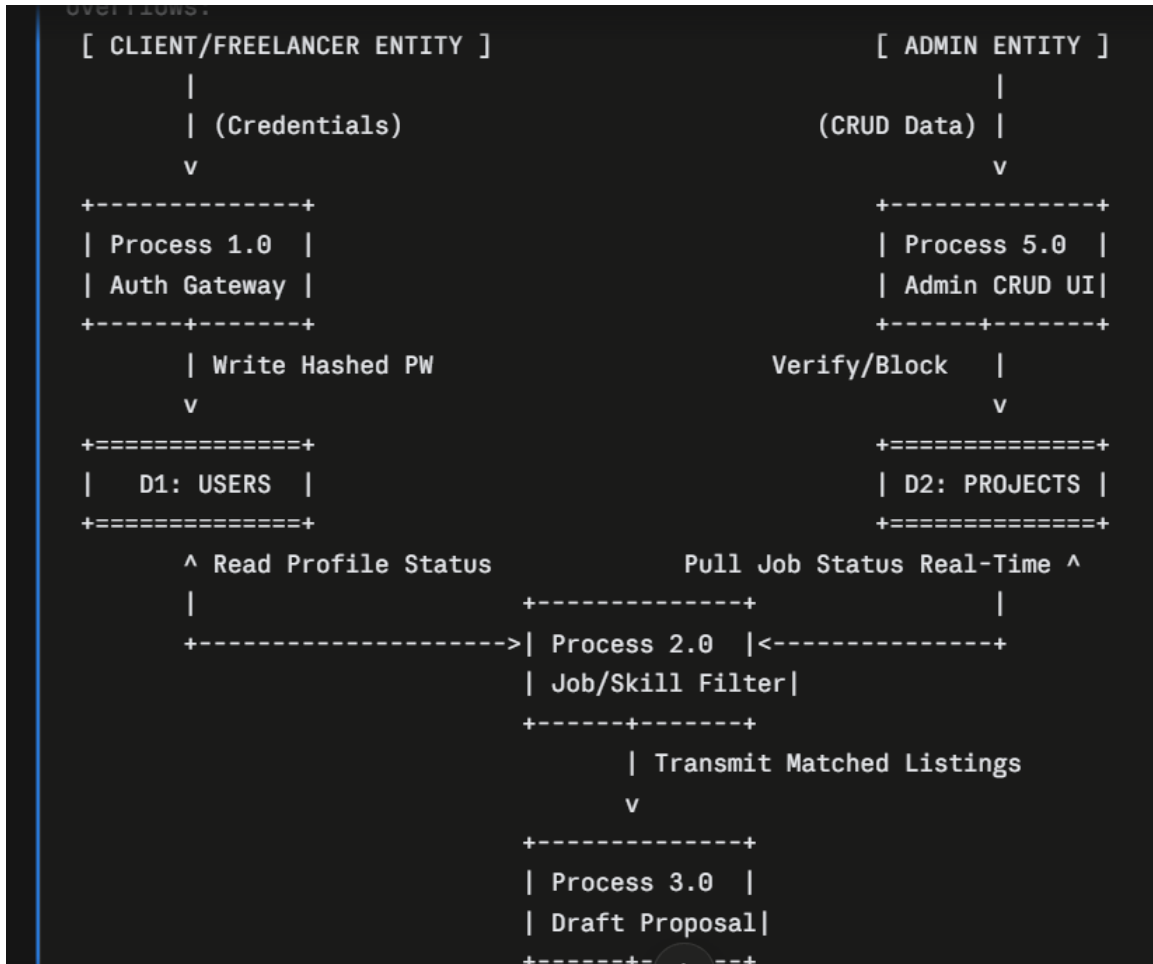
Date	31 January 2025
Team ID	
Project Name	FreelanceHub — Freelancing Marketplace
Maximum Marks	4 Marks

1. DFD Level 0: Context Diagram Diagrammatic Flow The Context Diagram maps out the global interactive scope of the FreelanceHub platform, establishing boundary sequences and dynamic inputs/outputs with your system core:



2. DFD Level 1: Operational Component Pipeline

The optimized Level 1 structural representation isolates your decoupled MERN stack architecture into compact transaction workflows designed to prevent line wrap page overflows:



3. Data Flow Diagram Entity and Process Dictionary

Element Identifier	Typology	Description & System Interactivity	Technology Mappings
Client / Freelancer Profile	External Entity	Initiates profile registrations, passes credentials, drafts job posts or proposals, and reads historic project/proposal tracking records.	React Frontend SPA
Platform Administrator	External Entity	Reviews system dashboard analytics, verifies/blocks users, and resolves transaction disputes.	Secured Admin Interface
Process 1.0	System Process	Processes inbound authorization payloads and signs cryptographically secure bearer JWT strings.	Express Middleware Token Pipeline

Process 2.0	System Process	Processes dynamic job/skill listings (Web Dev, Design, Data Science, etc.) via backend data aggregators.	Axios Endpoint Broker Nodes
Process 3.0	System Process	Maintains client-side draft proposal state and synchronizes bid amount computations before submission.	React Stateful Client Hooks
Process 4.0	System Process	Validates project availability, transitions project status to in-progress, and appends hire/assignment metadata.	Node.js REST API Subtrees
Process 5.0	System Process	Grants platform administrator personas full moderation control to verify, block, or review users.	Admin-Protected Route Gateways
D1: Users Collection	Data Store	Persists encrypted credentials, role tiers (client/freelancer/admin), and registration profile blocks.	Mongoose Object Document Model
D2: Projects Collection	Data Store	Retains real-time job status, skill/category tags, budget summaries, and posting metadata.	MongoDB Database Cluster
D3: Proposals Collection	Data Store	Houses verified bid records, cover letters, hire decisions, and status tracking updates.	MongoDB Persistence Tier